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Integrated broadband cable telecommunication networks (CABLE); Fourth generation transmission systems for interactive cable television services - IP cable modems; Part 1: General; DOCSIS® 3.1

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Foreword

This ETSI Standard (ES) has been produced by ETSI Technical Committee Integrated broadband cable telecommunication networks (CABLE).

The present document is part 1 of a multi-part deliverable covering the fourth generation transmission systems for interactive cable television services - IP cable modems, as identified below:

Part 1: "General; DOCSIS® 3.1";

Part 2: "Physical layer; DOCSIS® 3.1 [ANSI/SCTE 220-1 2016]";

Part 3: "MAC and upper layer protocols interface; DOCSIS® 3.1 [ANSI/SCTE 220-2 2016]";

Part 4: "Cable modem operations support system interface; DOCSIS® 3.1 [ANSI/SCTE 220-3 2016]";

Part 5: "Converged cable access platform operations support system interface; DOCSIS® 3.1 [ANSI/SCTE 220-4 2016]";

Part 6: "Security; DOCSIS® 3.1 [ANSI/SCTE 220-5 2016]".

This multi-part deliverable is based on the CableLabs DOCSIS® set of specifications and endorses the corresponding ANSI/SCTE Standards standardized in the United States by SCTE. Table 1 indicates for the specifications in this multi-part deliverable the endorsed ANSI/SCTE Standard and the corresponding CableLabs DOCSIS® specifications.

Table 1

ETSI Standards	ANSI/SCTE Standards	CableLabs DOCSIS® Specifications
ETSI ES 203 311-1	None	None
ETSI ES 203 311-2 [1]	ANSI/SCTE 220-1 2016	CM-SP-PHYv3.1-I08-151210
ETSI ES 203 311-3 [2]	ANSI/SCTE 220-2 2016	CM-SP-MULPIv3.1-I08-151210
ETSI ES 203 311-4 [3]	ANSI/SCTE 220-3 2016	CM-SP-CM-OSSIv3.1-I06-151210
ETSI ES 203 311-5 [4]	ANSI/SCTE 220-4 2016	CM-SP-CCAP-OSSIv3.1-I06-151210
ETSI ES 203 311-6 [5]	ANSI/SCTE 220-5 2016	CM-SP-SECv3.1-I05-151210

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

1 Scope

The present document is part of a series of specifications that defines the fourth generation of high-speed data-over-cable systems, commonly referred to as the DOCSIS 3.1 specifications. The standard was developed for the benefit of the cable industry, and includes contributions by operators and vendors from North and South America, Europe and Asia.

This generation of the DOCSIS specifications builds upon the previous generations of DOCSIS specifications (commonly referred to as the DOCSIS 3.0 and earlier specifications), leveraging the existing Media Access Control (MAC) and Physical (PHY) layers, but with the addition of a new PHY layer designed to improve spectral efficiency and provide better scaling for larger bandwidths (and appropriate updates to the MAC and management layers to support the new PHY layer). It includes backward compatibility for the existing PHY layers in order to enable a seamless migration to the new technology.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference/.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI ES 203 311-2: "Integrated broadband cable telecommunication networks (CABLE); Fourth generation transmission systems for interactive cable television services - IP cable modems; Part 2: Physical layer; DOCSIS® 3.1 [ANSI/SCTE 220-1 2016]".

[2] ETSI ES 203 311-3: "Integrated broadband cable telecommunication networks (CABLE); Fourth generation transmission systems for interactive cable television services - IP cable modems; Part 3: MAC and upper layer protocols interface; DOCSIS® 3.1 [ANSI/SCTE 220-2 2016]".

[3] ETSI ES 203 311-4: "Integrated broadband cable telecommunication networks (CABLE); Fourth generation transmission systems for interactive cable television services - IP cable modems; Part 4: Cable modem operations support system interface; DOCSIS® 3.1 [ANSI/SCTE 220-3 2016]".

[4] ETSI ES 203 311-5: "Integrated broadband cable telecommunication networks (CABLE); Fourth generation transmission systems for interactive cable television services - IP cable modems; Part 5: Converged cable access platform operations support system interface; DOCSIS® 3.1 [ANSI/SCTE 220-4 2016]".

[5] ETSI ES 203 311-6: "Integrated broadband cable telecommunication networks (CABLE); Fourth generation transmission systems for interactive cable television services - IP cable modems; Part 6: Security; DOCSIS® 3.1 [ANSI/SCTE 220-5 2016]".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

Not applicable.

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

cable modem: modulator-demodulator at the subscriber premises intended for use in conveying data communication on a cable television system

cable modem termination system: device located at the cable television system headend or distribution hub, which provides complementary functionality to the cable modems to enable data connectivity to a wide-area network

converged cable access platform: device located at the cable television system headend or distribution hub that combines the functionality of a cable modem termination system with that of an Edge QAM, providing high-density services to cable subscribers

distribution hub: facility in a cable network which performs the functions of a headend for customers in their immediate area, and which receives some or all of its content for transmission from a master headend in the same metropolitan or regional area

Edge QAM (EQAM): device that receives packets of digital video or data, repacketizes the video or data into an MPEG transport stream and digitally modulates the transport stream onto a downstream RF carrier using quadrature amplitude modulation (QAM)

headend: central facility that is used for receiving, processing and combining broadcast, narrowcast and other signals to be carried on a cable network

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

CCAP	Converged Cable Access Platform
CM	Cable Modem
CMTS	Cable Modem Termination System
CPE	Customer Premise Equipment
DHCP	Dynamic Host Configuration Protocol
EQAM	Edge QAM
HFC	Hybrid Fibre Coax
IP	Internet Protocol
IPDR	Internet Protocol Detail Record
IPv4	Internet Protocol version 4

IPv6	Internet Protocol version 6
LAN	Local Area Network
MAC	Media Access Control
PHY	Physical Layer
SCTE	Society of Cable Telecommunications Engineers
SNMP	Simple Network Management Protocol

4 Background

4.1 Broadband Access Network

A coaxial-based broadband access network is assumed. This may take the form of either an all-coax or hybrid-fibre/coax (HFC) network. The generic term "cable network" is used in the present document to cover all cases.

A cable network uses a tree-and-branch architecture with analogue transmission. The key functional characteristics assumed in the present document are the following:

- Two-way transmission.
- A maximum optical/electrical spacing between the CMTS and the most distant CM of 160 km in each direction, although typical maximum separation may be 15 - 25 km.

At a propagation velocity in fibre of approximately 5 ns/m, 160 km of fibre in each direction results in a round-trip delay of approximately 1,6 ms.

4.2 DOCSIS Network and System Architecture

The elements that participate in the provisioning of DOCSIS services are shown in figure 1.

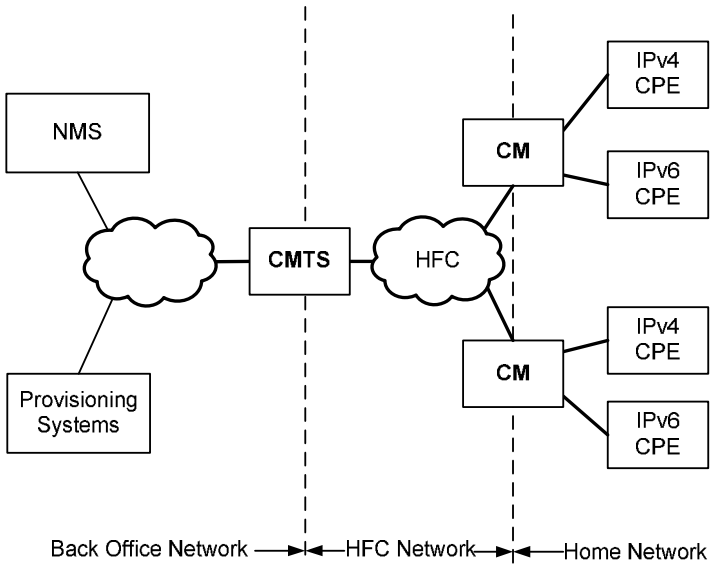


Figure 1: The DOCSIS Network

The CM connects to the operator's cable network and to a home network, bridging packets between them. Many CPE devices can connect to the CM's LAN interfaces. CPE devices can be embedded with the CM in a single device, or they can be separate, standalone devices (as shown in figure 1). CPE devices may use IPv4, IPv6, or both forms of IP addressing. Examples of typical CPE devices are gateways, home routers, set-top devices, personal computers, etc.

The CMTS connects the operator's back office and core network with the cable network. Its main function is to forward packets between these two domains, and between upstream and downstream channels on the cable network.

Various applications are used in the back office to provide configuration and other support to the devices on the DOCSIS network. These applications use IPv4 and/or IPv6, as appropriate to the particular operator's deployment. Applications include:

Provisioning Systems:

- The DHCP servers provide the CM with initial configuration information, including IP address(es), when the CM boots.
- The Config File server is used to download configuration files to CMs when they boot. Configuration files are in binary format and permit the configuration of the CM's parameters.
- The Software Download server is used to download software upgrades to the CM.
- The Time Protocol server provides time protocol clients, typically CMs, with the current time of day.
- The Certificate Revocation server provides certificate status.

Network Management System (NMS):

- The SNMP Manager allows the cable operator to configure and monitor SNMP Agents, typically the CM and the CMTS.
- The Syslog server collects messages pertaining to the operation of devices.
- The IPDR Collector server allows the operator to collect bulk statistics in an efficient manner.

4.3 Service Goals

As cable operators have widely deployed high-speed data services on cable television systems, the demand for bandwidth has increased. To this end, it was decided to add new features to the DOCSIS specification for the purpose of increasing system and channel capacity, increasing peak speeds, improving scalability, enhancing network security and network maintenance practices and deploying new service offerings.

The DOCSIS system allows transparent bidirectional transfer of Internet Protocol (IP) traffic, between the cable system headend and customer locations, over an all-coaxial or hybrid-fibre/coax (HFC) cable network. The flow of traffic is shown in simplified form in figure 2.

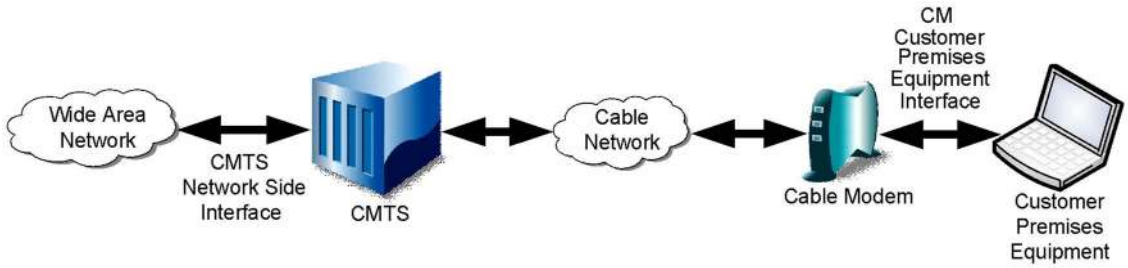


Figure 2: Transparent IP Traffic Through the Data-Over-Cable System

4.4 Backward Compatibility

The present document defines the DOCSIS 3.1 interface. Prior generations of DOCSIS were commonly referred to as the DOCSIS 1.0, 1.1, 2.0 and 3.0 interfaces. DOCSIS 3.1 provides backward-compatibility with equipment built to certain previous versions. DOCSIS 3.1-compliant CMs interoperate seamlessly with DOCSIS 3.1 and DOCSIS 3.0 CMTSs. DOCSIS 3.1-compliant CMTSs seamlessly support DOCSIS 3.0, DOCSIS 2.0, and DOCSIS 1.1 CMs.

4.5 Reference Architecture

The reference architecture for data-over-cable services and interfaces is shown in figure 3.

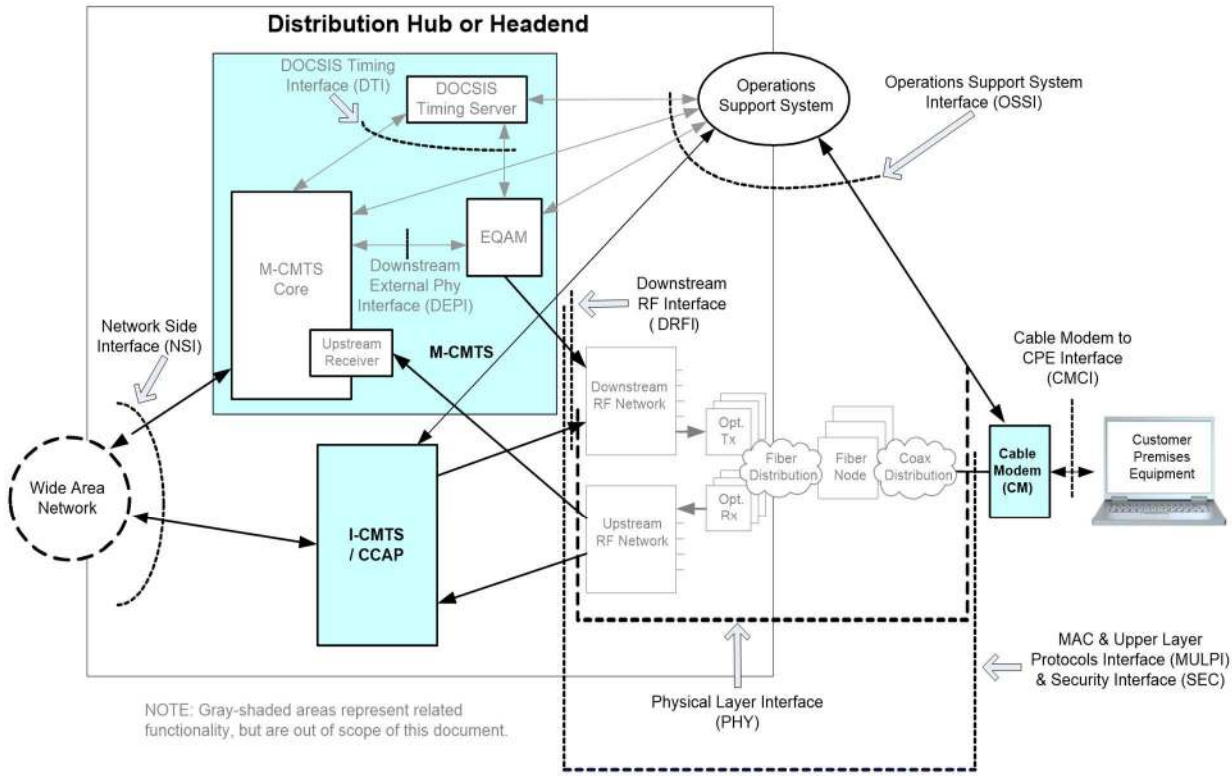


Figure 3: Data-over-Cable Reference Architecture

The lighter shaded areas are related functionality, but are out of the scope of DOCSIS specifications. Boxes represent functional components and arrows represent interfaces. Interfaces are identified using dashed lines.

5 Overview of the multi-part ETSI Standard

5.1 Part 1: General; DOCSIS® 3.1

The present document.

5.2 Part 2: Physical layer; DOCSIS® 3.1

ETSI ES 203 311-2 [1] defines the interface for the physical layer of DOCSIS 3.1.

There are differences in the cable spectrum planning practices adopted for different networks in the world. For the new PHY layer defined in the present document, there is flexibility to deploy the technology in any spectrum plan; therefore, no special accommodation for different regions of the world is required for this new PHY layer.

However, due to the inclusion of the DOCSIS 3.0 PHY layers for backward compatibility purposes, there is still a need for different region-specific physical layer technologies. Therefore, three options for physical layer technologies are included in the standard, which have equal priority and are not required to be interoperable. One technology option is based on the downstream channel identification plan that is deployed in North America using 6 MHz spacing. The second technology option is based on the corresponding European multi-program television distribution. The third technology option is based on the corresponding Chinese multi-program television distribution. All three options have the same status, notwithstanding that the document structure does not reflect this equal priority. It is not required that equipment built to one option interoperates with equipment built to the other.

5.3 Part 3: MAC and upper layer protocols interface; DOCSIS® 3.1

ETSI ES 203 311-3 [2] defines the MAC layer protocols of DOCSIS 3.1 as well as requirements for upper layer protocols (e.g. IP, DHCP, etc.).

5.4 Part 4: Cable modem operations support system interface; DOCSIS® 3.1

ETSI ES 203 311-4 [3] defines the Operations Support System Interface (OSSI) requirements for the Cable Modem (CM).

5.5 Part 5: Converged cable access platform operations support system interface; DOCSIS® 3.1

ETSI ES 203 311-5 [4] defines the requirements necessary for the configuration, fault management and performance management of Cable Modem Termination Systems (CMTS) and the Converged Cable Access Platform (CCAP) system. The intent is to define a common, cross-vendor set of functionality for the configuration and management of CMTSs and CCAPs.

Part 5 defines a configuration object model for the configuration of the CCAP. It also defines the SNMP management requirements for a CCAP. These SNMP requirements include both protocol conformance and management object definitions, based largely upon existing industry standard management objects found in DOCSIS CMTSs and Universal EQAMs. In addition, it defines the event messaging requirements of a CCAP system.

5.6 Part 6: Security; DOCSIS® 3.1

ETSI ES 203 311-6 [5] defines the security requirements of DOCSIS 3.1.

6 Requirements for Compliance

To claim compliance to ETSI ES 203 311, implementations shall comply with the requirements statements in all parts of the present multi-part deliverable.

Annex A (informative): Change History

Date	Version	Information about changes
2019	1.1.1	First publication of the document after approval by ETSI TC CABLE

History

Document history		
V1.1.1	March 2019	Membership Approval Procedure MV 20190513: 2019-03-14 to 2019-05-13
V1.1.1	May 2019	Publication